

Threat Hunt 001: Suspicious Parent-Child Process Chains

Status: Complete — CONFIRMED **Analyst:** Detection Engineering **Date:** 2026-04-04 **Dataset:** EVTX-ATTACK-SAMPLES (278 EVTX files, 37,364 total events, 1,501 Sysmon EventID 1 records)

Hypothesis

An adversary operating in the sampled environment will execute a shell or scripting interpreter as a direct child of an unusual parent (Office suite, spooler, WMI provider host), which should be detectable in Sysmon Process Create (EventID 1) records.

Data Sources

- data/processed/evtx-json/**/* .jsonl (all 278 files)
- Sysmon EventID 1 (Process Create)
- Fields: `Event.EventData.ParentImage`, `Event.EventData.Image`, `Event.EventData.CommandLine`

Method

1. Extracted all 1,501 Sysmon EventID 1 events into a flat JSONL using `jq`.
2. Filtered for parents matching the suspicious list: `WINWORD|EXCEL|POWERPNT|OUTLOOK|spoolsv|WmiPrvSE|wmiprvse`.
3. Built baseline distributions of top parent and child processes to compare findings against expected system activity.

Evidence file: `hunts/hunt_001_evidence/all_proc_creation.jsonl`

Findings

Finding 1 — Masqueraded WINWORD.exe dropping and cleaning up a DLL (T1204.002, T1059.003)

Parent-child chain observed:

```
C:\Users\IEUser\AppData\Roaming\WINWORD.exe -> C:\Windows\SysWOW64\cmd.exe
CommandLine: cmd /c ping 127.0.0.1&&del del /F /Q /A:H "C:\Users\IEUser\AppData\Roaming\wwlib.dll"
```

- The Microsoft Word binary is running from AppData\Roaming\, not Program Files\Microsoft Office\.
- It spawns cmd.exe with an obfuscated delete command targeting wwlib.dll.
- ping 127.0.0.1 is a classic **sleep replacement** used to delay execution in sandboxes and simple macros.
- The del /F /Q /A:H flag combination removes a **hidden** DLL, consistent with an implant leaving no trace.

Finding 2 — WMI-driven lateral movement (T1047, T1021.006)

Multiple WmiPrvSE.exe process creations:

```
WmiPrvSE.exe -> cmd.exe /Q /c cd 1> \\127.0.0.1\ADMIN$
\__1556656369.7 2>&1
WmiPrvSE.exe -> cmd.exe /Q /c whoami /all 1> \\127.0.0.1\ADMIN$
\__1556656369.7 2>&1
WmiPrvSE.exe -> cmd.exe /Q /c cd \ 1> \\127.0.0.1\C$\WqEVwJZY0e
2>&1
```

- The /Q flag (quiet) plus output redirection to \\127.0.0.1\ADMIN\$ __<epoch> is the **Impacket wmiexec.py signature**.
- The files __1556656369.7 (epoch-timestamped) and WqEVwJZY0e (random 10-char) are characteristic of the tool's temp output.
- whoami /all is immediate post-exploitation reconnaissance.
- Technique mapping: T1047 (WMI), T1021.006 (WinRM/remote exec), T1033 (System Owner/User Discovery).

Finding 3 — LSASS credential dumping via comsvcs.dll (T1003.001)

```
WmiPrvSE.exe -> rundll32.exe  
CommandLine: rundll32 C:\windows\system32\comsvcs.dll, MiniDump  
4868 C:\Windows\System32\notepad.bin full
```

- **Validates detection rule** `win_comsvcs_minidump.yml`.
- 4868 is the LSASS PID at the time of the dump.
- `notepad.bin` in `System32\` is a hardcoded output path used by public PoCs.
- The `full` argument dumps the complete process memory, capturing credential material.

Finding 4 — Execution from Volume Shadow Copy (T1564.002, T1055)

```
WmiPrvSE.exe ->  
\Device\HarddiskVolumeShadowCopy7\Windows\Temp\svhost64.exe  
CommandLine: \\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy7\  
\Windows\Temp\svhost64.exe
```

- Binary executed directly from a shadow copy, bypassing normal path-based controls.
- `svhost64.exe` masquerades as the legitimate `svchost.exe` but with a 6/4 twist.
- `\\?\GLOBALROOT\Device\...` prefix is a common way to access shadow copies without mounting.

Baseline Comparison

Top parent processes across 1,501 events:

Parent	Count
<code>cmd.exe</code>	547
<code>powershell.exe</code>	232
<code>services.exe</code>	137
<code>svchost.exe</code>	111
<code>explorer.exe</code>	52
<code>winlogon.exe</code>	39

Parent	Count
WmiPrvSE.exe	14

WmiPrvSE.exe at 14 occurrences is low-volume compared to the baseline. All 14 are suspicious — none reflect routine OS activity in this corpus.

Conclusion

CONFIRMED — the hypothesis holds. Multiple independent adversarial behaviors are observable in the Sysmon Process Create stream:

1. **Macro-based initial execution** masquerading as Office (Finding 1).
2. **WMI-based lateral movement** with Impacket signatures (Finding 2).
3. **Credential dumping** via comsvcs.dll (Finding 3) — this validates our existing detection rule.
4. **Shadow-copy execution** for defense evasion (Finding 4).

New Detections Enabled

Findings 1, 2, and 4 are **not yet covered** by the current 15-rule library. The following rules should be added:

Rule Name	Technique	Basis
win_office_masquerade_appdata.yml	T1036.005	Office binary running from AppData
win_wmi_impacket_redirect.yml	T1047	WmiPrvSE -> cmd.exe with ADMIN\$/C\$ re \\?
win_shadow_copy_execution.yml	T1564.002	\GLOBALROOT\Device\HarddiskVolumeS in CommandLine

Finding 3 is already covered by win_comsvcs_minidump.yml.

ATT&CK Mapping

- T1204.002 — User Execution: Malicious File
- T1059.003 — Windows Command Shell
- T1036.005 — Masquerading: Match Legitimate Name or Location
- T1047 — Windows Management Instrumentation

- T1021.006 — Remote Services: Windows Remote Management
- T1033 — System Owner/User Discovery
- T1003.001 — OS Credential Dumping: LSASS Memory
- T1564.002 — Hide Artifacts: Hidden Users